Week 9, Lecture 1: From Optimization to MCMC

Instructor notes. Part 1: prep reading. Part 2: podium cheat sheet

How to prepare (for students)

- **Big picture.** Last Wednesday we opened up `optim()` and saw how Newton's method uses the slope and curvature of the NLL to step toward the MLE. Today: two search methods that need no derivatives (Nelder-Mead and simulated annealing), then a small change to simulated annealing that turns it from a way to *find the best* parameter values into a way to *sample the whole posterior distribution*. That method, MCMC, is how most Bayesian models are fit.
- **Read:** Bolker 7.2.3.2 and 7.2.4 (Nelder-Mead and simulated annealing, printed p. 302-310) and 7.3 through 7.3.2 (MCMC, burn-in, convergence, p. 310-315).
- **Review:** Bayes' rule for parameters (posterior $\propto$ likelihood $\times$ prior) and the Beta-binomial germination example from Week 8; the log rule $\log(a/b) = \log a - \log b$.
- **Tuesday's lab** has you build and test your own MCMC sampler, so bring questions from today.

Overview (instructor only)

As taught (Mon., Oct. 19). The closing discussion (three germination chains with different step sizes) wasn't reached. Lab 7 Act 1 Part 2 asked students to judge chains first, so the discussion became a short debrief at the start of Wednesday (Week 9 Lecture 2, section 0) and was removed from these notes.

Beat	Mode	Time
1. Big picture	BOARD	~4 min
2. Nelder-Mead	BOARD $\rightarrow$ R	~7 min
3. Simulated annealing	BOARD $\rightarrow$ R	~8 min
4. Metropolis	BOARD $\rightarrow$ R	~15 min
5. Wrap	BOARD	~2 min

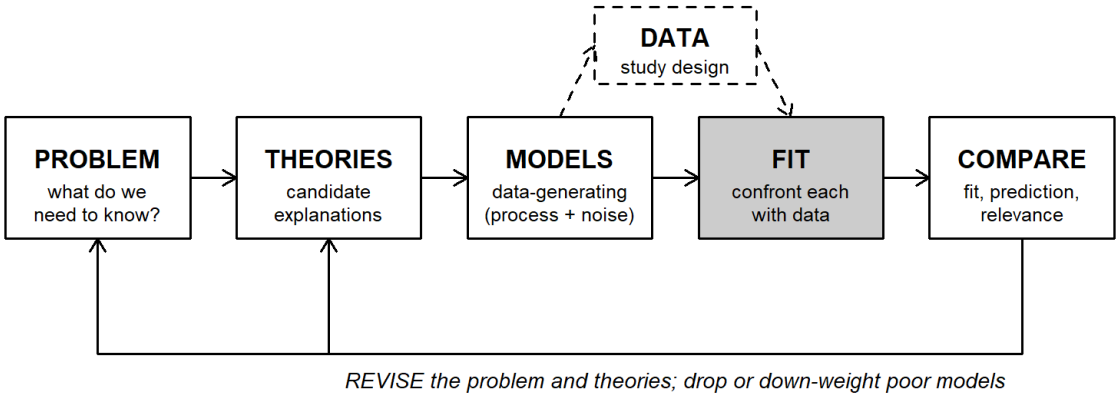
No WOD today, to leave room so Tuesday's lab can start without a long lecture. **BOARD** = whiteboard; **R** = switch to `CODE_Week9_MCMCAndStan.R` (Monday section; numbers match). The script holds only the R demos; figures in these notes are board references only.

Where this came from. Sections 2-3 are the Nelder-Mead and simulated annealing material planned for Week 8 Lecture 2 (Oct. 14), which only got through Newton's method. Both are trimmed to their core ideas. The shape-scale correlation pause and the reparameterization "if time" material from Week 8 become Lab 7 Act 2's centering step, where removing a strong parameter correlation fixes a badly mixing chain (the same reason the myxo banana mixes slowly).

If running behind, skip the SANN pause in section 3 and the two-parameter myxo chain in section 4.

Part 1: Lecture content

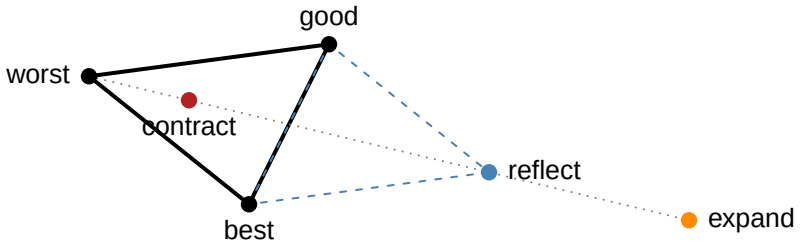
1. Big picture (~4 min) [BOARD]



- Workflow on the board (a quick version of the above); point at **FIT**.
- Recap: fitting = finding the bottom of the NLL surface. Newton and BFGS use slopes; fast near the answer, can fail far from it.
- Today, two questions: (1) how do we search when slopes are unavailable or misleading? (2) For Bayesian fits, how do we get the posterior when there are too many parameters for a grid and no conjugate shortcut?
- The answer to (2) grows out of the answer to (1).

2. Nelder-Mead (~7 min) [BOARD → R]

Board. Sketch one triangle (three points for two parameters) and its moves:



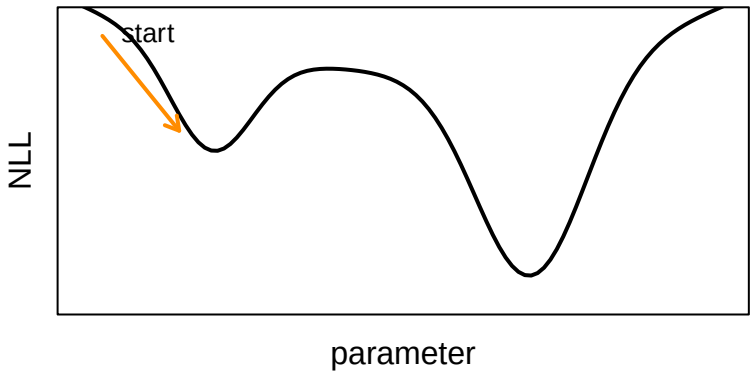
- Reflect the worst point through the midpoint of the other two. Best yet? Expand. Still bad? Contract. Nothing works? Shrink toward the best point.
- No derivatives: only compares NLL values.

R demo (myxo Gamma surface from Week 8). The triangle expands across the contours, contracts as it turns into the valley, then crawls along the valley floor. `optim()`'s default is Nelder-Mead: 147 NLL evaluations, NLL 37.667.

Point to land. Slower than BFGS, but bumps, kinks, and thresholds in the NLL don't fool it.

3. Simulated annealing (~8 min) [BOARD → R]

Board. A curve with two valleys (Bolker's Figure 7.1). Any downhill method finds whichever valley it starts in.



The rule (Bolker p. 306):

1. Propose a random nearby point.
2. Lower NLL: move there.
3. Higher by Δ: move there anyway with probability $e^{-\Delta/T}$.
4. Repeat; lower the "temperature" T now and then.

Hot: accepts most uphill steps and explores. Cold: almost pure downhill. Start hot, cool slowly.

R demo. Acceptance table, then the annealer on the myxo surface: it finds the valley, wanders along it, settles as T falls. About 3,000 evaluations for what Nelder-Mead did in 147.

Pause (~1 min, skippable). `optim(..., method = "SANN")` with defaults gives NLL 40.77 after 10,000 evaluations and `convergence = 0` (Nelder-Mead: 37.67). Broken? *Land:* untuned, not broken. SANN always reports 0. Use a global search to find the right valley, then polish with Nelder-Mead or BFGS.

4. Metropolis (~15 min) [BOARD → R]

Key idea. Keep the annealing rule but change three things, and the chain of visited points becomes a sample from the posterior.

Board, in three steps.

1. Fix $T = 1$ and never cool. The acceptance probability for an uphill step becomes $e^{-(\text{NLL}_{new} - \text{NLL}_{old})} = \mathcal{L}(\theta_{new})/\mathcal{L}(\theta_{old})$.
2. Swap the likelihood for the posterior:

$$P(\text{accept}) = \min\left(1, \frac{P(\theta_{new} \mid y)}{P(\theta_{old} \mid y)}\right)$$

3. Keep **every** step, not only the best one.

Then write Bayes' rule into the ratio:

$$\frac{P(\theta_{new} \mid y)}{P(\theta_{old} \mid y)} = \frac{\mathcal{L}(\theta_{new})\, P(\theta_{new})/P(y)}{\mathcal{L}(\theta_{old})\, P(\theta_{old})/P(y)}$$

$P(y)$ cancels. Computing it is why Week 7 used a grid and Week 8 relied on conjugacy; neither works with many parameters or arbitrary priors. Metropolis never needs it.

Why it lands on the posterior (board, 1 min). Two values, A and B, with $P(A \mid y) = 2\,P(B \mid y)$. From B, a proposal to A is always accepted; from A, a proposal to B is accepted half the time. In the long run the chain sits at A twice as often as at B. Same logic everywhere: time spent near $\theta \propto$ posterior.

Sticking point. A rejected proposal means *record the current value again.* Students often skip rejected steps; that throws away the extra weight the chain puts on high-posterior regions. Related: the chain is not supposed to converge to a point. Wandering is the goal.

R demo.

- Germination (30 of 40, flat prior; exact posterior Beta(31, 11)). Start at 0.5, proposal sd 0.1, 5,000 steps. Acceptance 59%. Chain mean 0.734 vs. exact 0.738; 95% interval 0.59-0.85 vs. exact 0.60-0.86. The histogram matches the curve we already know.
- Code note: compute log posteriors as sums of logs, never a raw product (27 densities multiplied together can underflow).

- **Pause (~1 min).** `anneal()` accepted with `runif(1) < exp(-delta / temp)`; `metropolis()` uses `log(runif(1)) < lp_prop - lp_current`. Different rules? *Land:* no. $u < e^x \iff \log u < x$, and with $T = 1$ and the NLL difference replaced by the log posterior difference, the two lines make the same decision. Only a *difference* is exponentiated, so neither can underflow.
- Myxo Gamma, two parameters, flat priors, 20,000 steps: the cloud of points fills the banana. 95% credible interval for shape: 29 to 81 (first 2,000 steps dropped; say why in one line, and leave burn-in for Lab 7 and Wednesday). This chain mixes slowly along the banana (other step sizes shift the lower bound by several units): shape and scale trade off, so most independent jumps leave the narrow ridge. Lab 7 Act 2 shows the same problem on a straighter ridge, and Wednesday shows a sampler that follows it.

5. Wrap (~2 min) [BOARD]

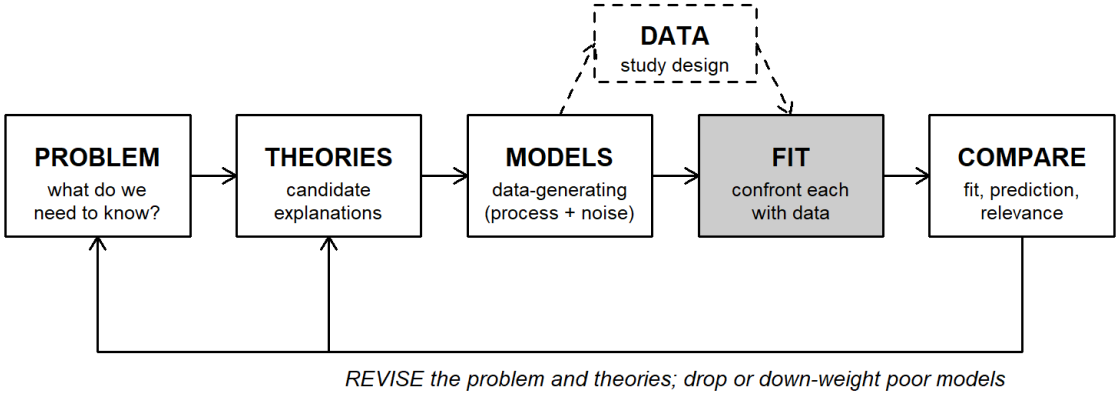
- Point at **FIT**. Grid, Newton/BFGS, Nelder-Mead, annealing: ways to find the minimum. Metropolis: a way to sample the posterior.
- Same engine (random proposal + acceptance rule); different goal.
- Tomorrow: you build one and learn how to tell whether to trust it.

PART 2: PODIUM CHEAT SHEET

Terse on purpose. Detail: Part 1. R = Monday section of CODE_Week9_MCMCAndStan.R.

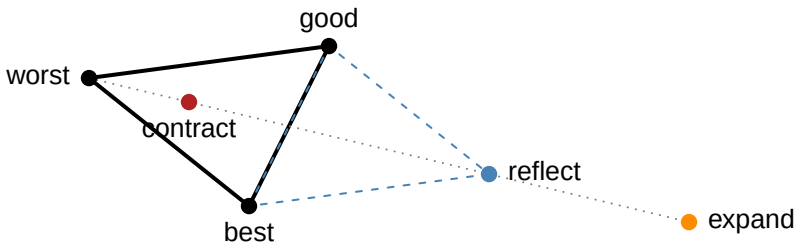
No WOD. Lecture ~36 min (discussion moved to Wed.). Behind? Skip SANN pause and myxo chain.

1. Big picture (~4 min) [BOARD]



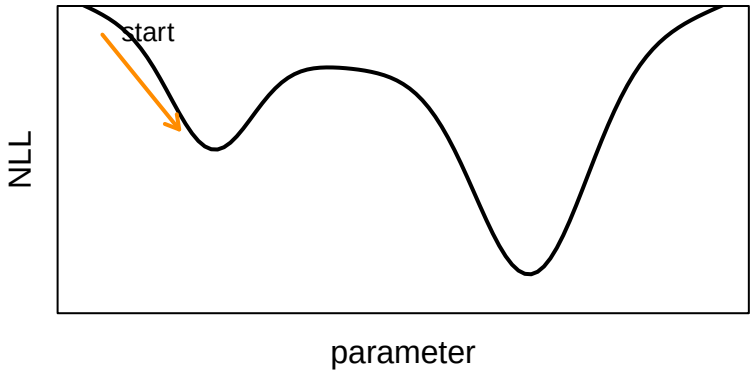
- Workflow on board; point at **FIT**
- Newton/BFGS: slopes; fast near answer, fail far away
- Q1: search without slopes?
- Q2: posterior with too many params for a grid, no conjugacy?
- Q2's answer grows out of Q1's

2. Nelder-Mead (~7 min) [BOARD → R]



- Reflect worst; best yet → expand; still bad → contract; else shrink
- Compares NLL values only, no derivatives
- **R:** triangles on myxo surface; `optim()` default, 147 evals
- Slower than BFGS; not fooled by bumps or kinks

3. Simulated annealing (~8 min) [BOARD → R]



- Downhill methods find the nearest valley
- Random step; worse by Δ: accept w.p. $e^{-\Delta/T}$; cool T
- Hot explores, cold settles
- **R:** acceptance table; annealer path; ~3,000 evals vs. 147

Pause (skippable): SANN: NLL 40.77, conv 0. Broken?

- Untuned; always 0. Find valley, then polish with NM/BFGS

4. Metropolis (~15 min) [BOARD → R]

1. $T = 1$: $e^{-\Delta \text{NLL}} = \mathcal{L}_{new}/\mathcal{L}_{old}$
2. Posterior instead: accept w.p. $\min\left(1, \frac{P(\theta_{new}|y)}{P(\theta_{old}|y)}\right)$
3. Keep **every** step

- Write Bayes in the ratio: $P(y)$ **cancels**
- A vs. B, A twice as probable: B→A always, A→B half ⇒ 2:1 time
- **Rejected = record current value again**
- Not converging to a point; wandering is the goal

R:

- Germination: mean 0.734 vs. 0.738; accept 59%
- Log posterior = sum of logs (a product underflows)
- **Pause:** `runif(1) < exp(-delta / temp)` vs. `log(runif(1)) < lp_prop - lp_current`? Same rule ($u < e^x \iff \log u < x$)
- Myxo: cloud fills banana; shape 29-81

5. Wrap (~2 min) [BOARD]

- **FIT**: optimizers find the minimum; Metropolis samples the posterior
- Same engine, different goal
- Tomorrow: build one, learn when to trust it